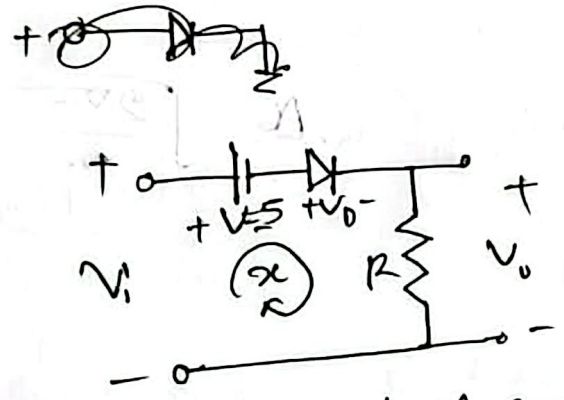
Series Clipper



KVL for +ve half-cycle.

$V_i(t) - V - V_D - V_o(t) = 0$

$\Rightarrow V_o(t) = V_i(t) - V - V_D$

$\Rightarrow V_{o_{peak}} = V_m - V - V_D$

for -ve half-cycle.

~~$V_i(t) = 0$~~

$V_o(t) = 0$

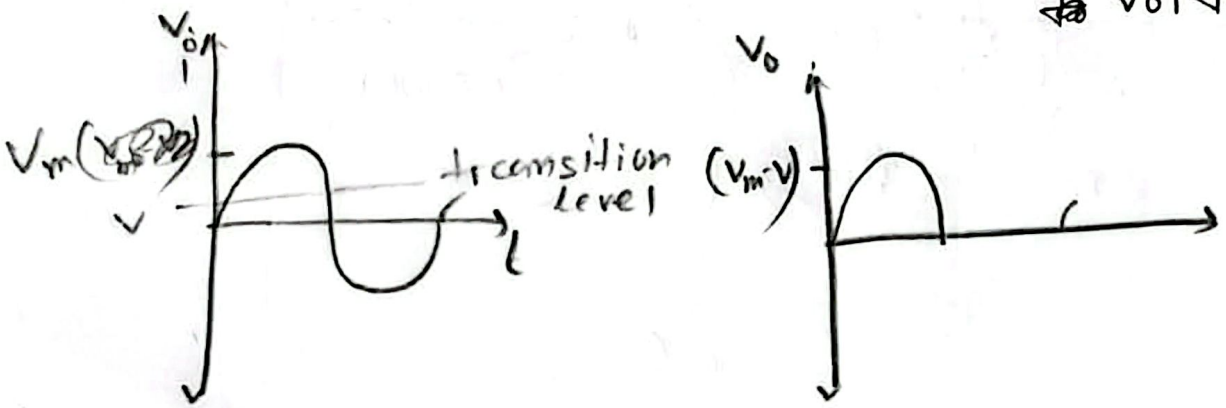
Transition voltage:

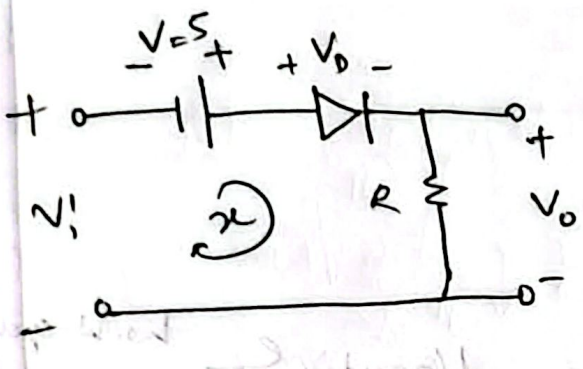
KVL in loop x,

$V_i(t) - V - V_D - V_o(t) = 0$

$\Rightarrow V_i(t) = V + V_D$ [at transition level voltage output is zero]

$\Rightarrow V_i(t) = V + V_D$ = transition ~~level~~ voltage





For positive half-cycle

$V_i(t) + V - V_D - V_o(t) = 0$

$\Rightarrow V_o(t) = V_i(t) + V - V_D$

$\Rightarrow V_{o_{peak}} = V_m + V - V_D$

Transition Voltage:

KVL in loop ex,

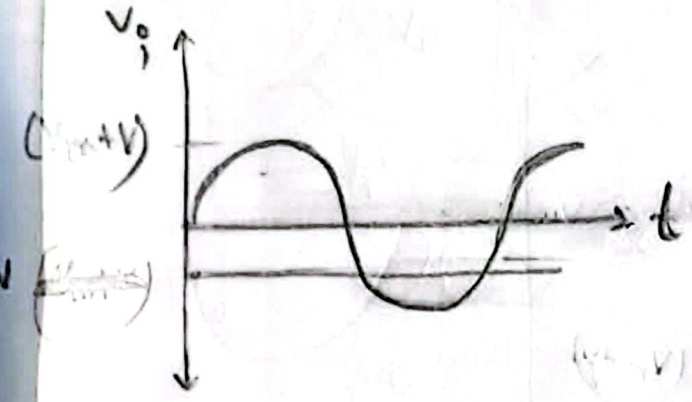
$V_i(t) + V - V_D = V_o(t)$

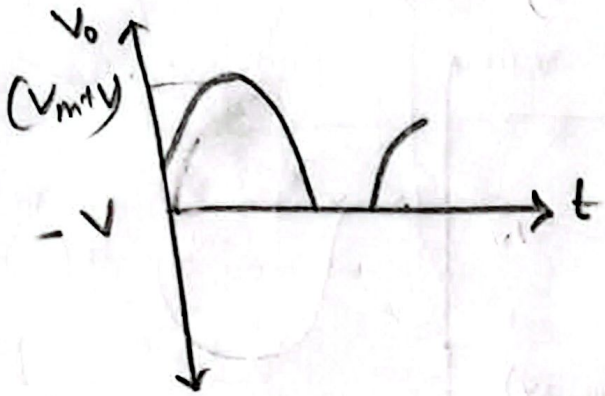
$\Rightarrow V_i(t) = -V + V_D$ [$V_o(t) = 0$ at t.l.]

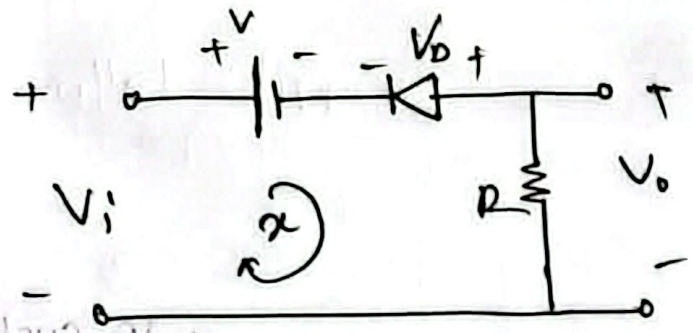
$\Rightarrow V_i(t) = -V + V_D$ = Transition voltage

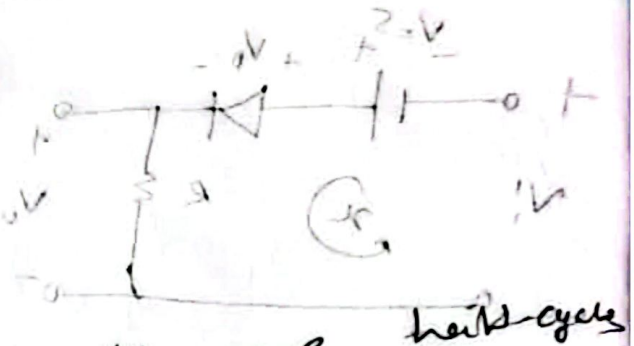
For negative half-cycle,

$V_i(t) = 0$









For positive half-cycle,

$V_o(t) = 0$

[Diode is reverse bias]

For Negative half-cycle

$V_i(t) = -V + V_D + V_o(t) = 0$

$\Rightarrow V_o(t) = V_i(t) - V + V_D$

$\Rightarrow V_{o\,peak} = -V_m - V + V_D$

[V_m -ve cause in (-ve) half-cycle]

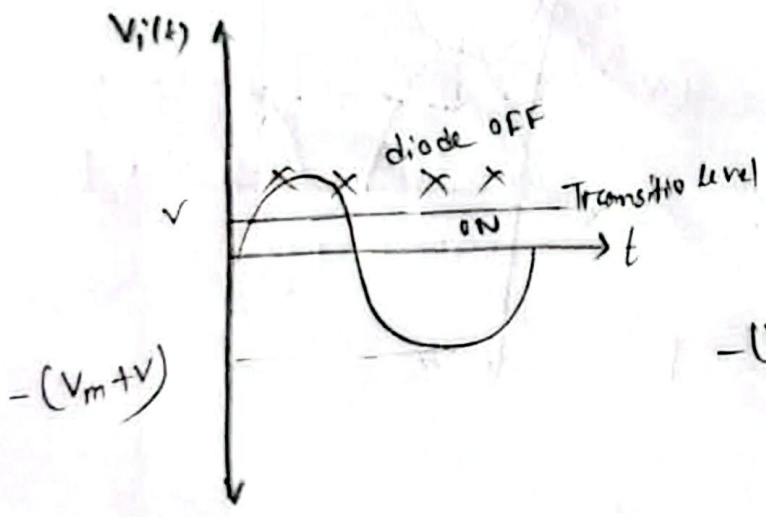
Transition voltage:

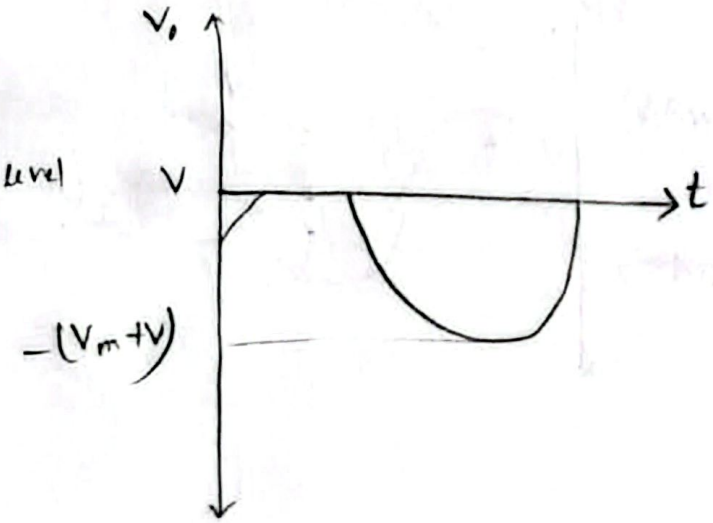
KVL in loop 2

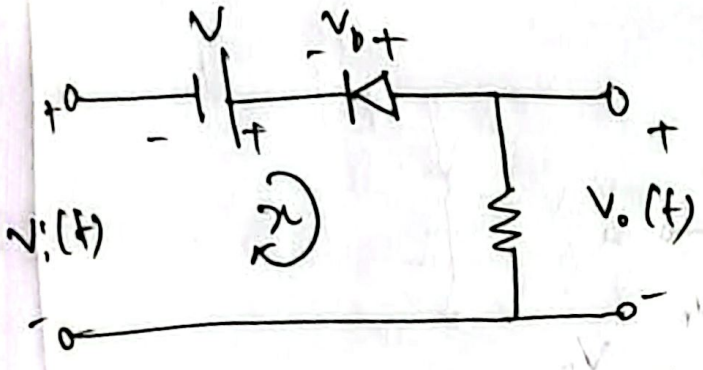
$V_i(t) - V + V_D = V_o(t)$

$\Rightarrow V_i(t) - V + V_D = 0$ [at t.v. $V_o(t) = 0$]

$\Rightarrow V_i(t) = V - V_D$ = Transition Voltage.







For positive half-cycle;

$v_o(t) = 0$

For Negative half-cycle;

$v_i(t) + V + V_D - V_o(t) = 0$

$\Rightarrow V_o(t) = v_i(t) + V + V_D$

$\Rightarrow V_{o\,peak} = -V_m + V + V_D$

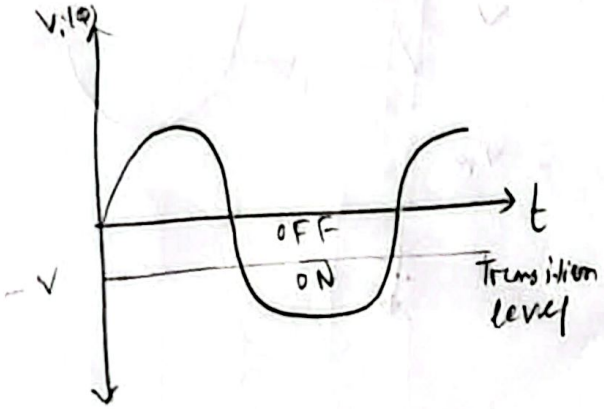
Transition voltage:

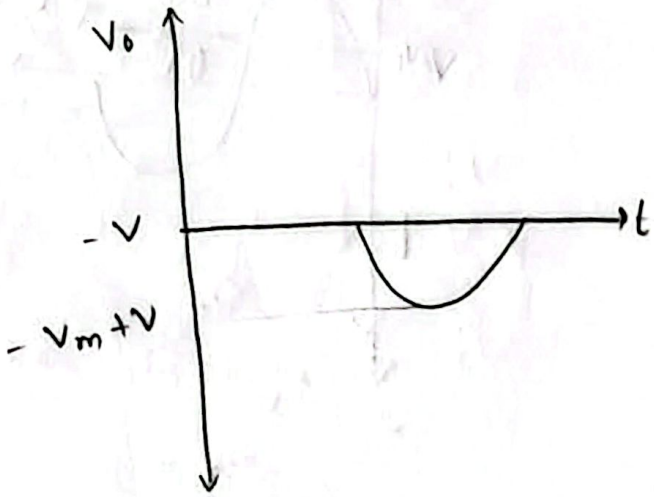
KVL in loop;

$v_i(t) + V + V_D = V_o(t)$

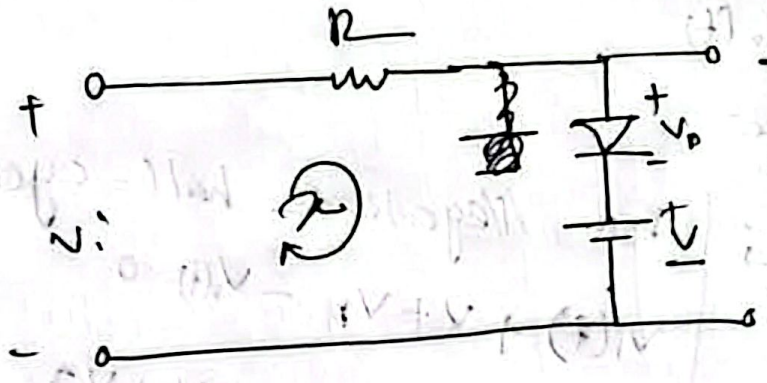
$\Rightarrow v_i(t) + V + V_D = 0$ [$\because V_o(t) = 0$]

$\Rightarrow v_i(t) = -V - V_D$ = Transition voltage





Parallel Clipper:



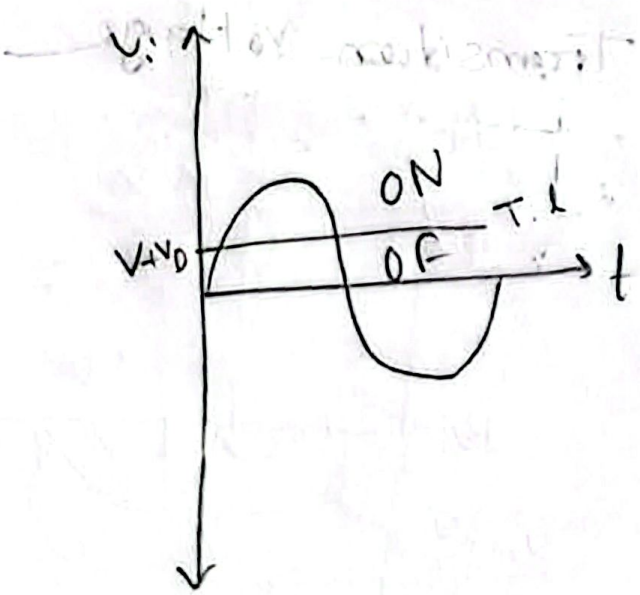
when, diode is on; $V_o = V + V_D$

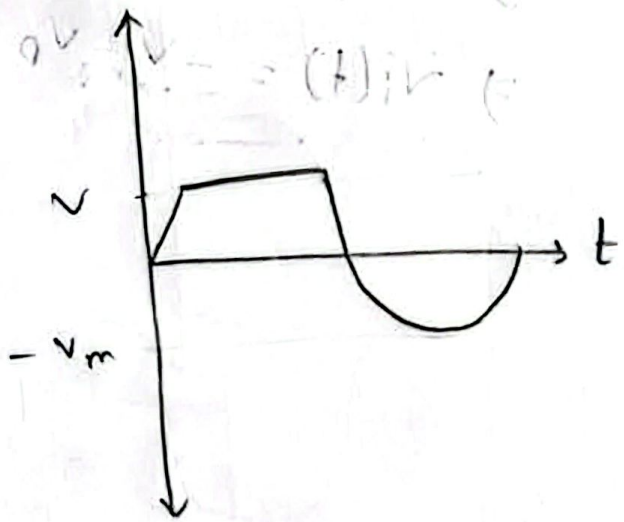
" " off, $V_o = V_i$

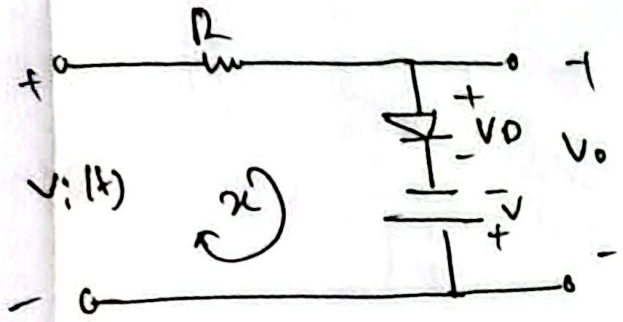
KVL in loop 2,

$V_i - V_D - V = 0$

$\Rightarrow V_i = V + V_D$ = Transition level







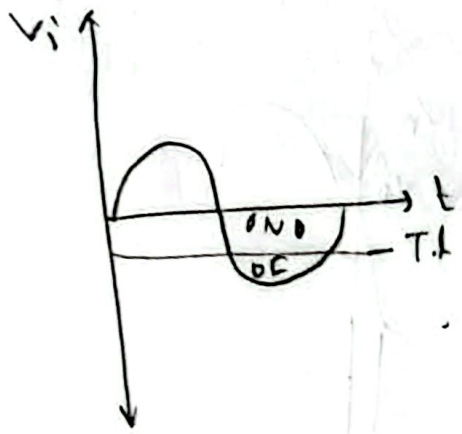
When, diode is ON, $V_o = -V + V_D$

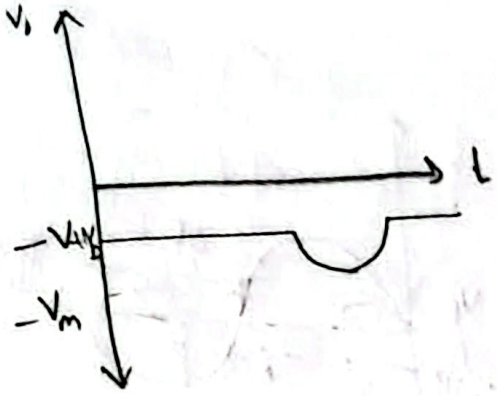
When, diode is OFF, $V_o = V_i$

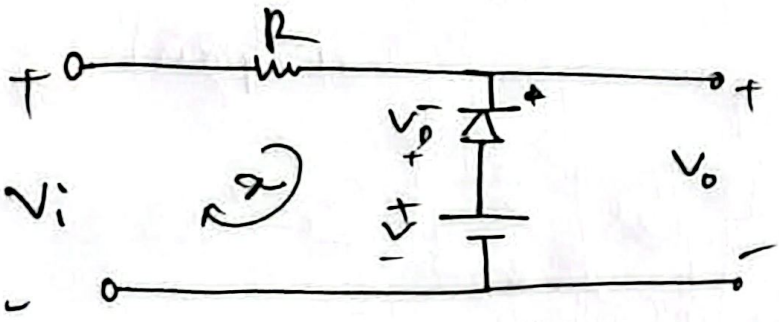
KVL in loop x,

$V_i - V_D + V = 0$

$\Rightarrow V_i = -V + V_D$ = Transition level







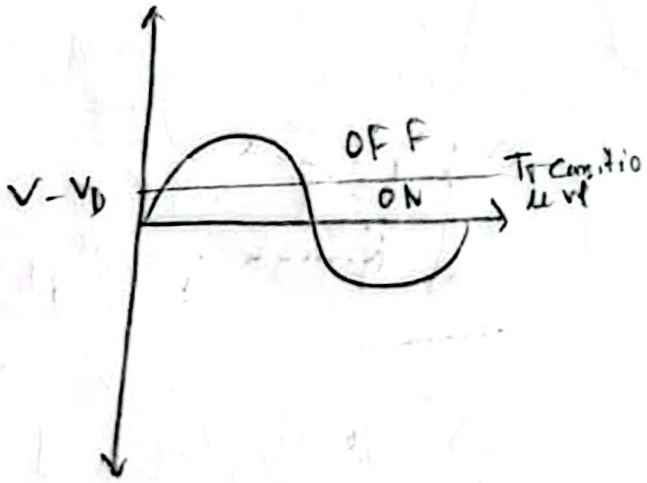
When, ~~loop is~~ diode is ON, $V_o = V_i - V_D$

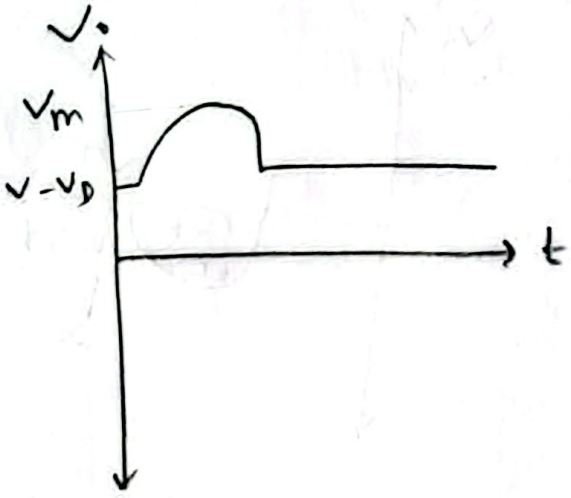
OFF, $V_o = V_i$

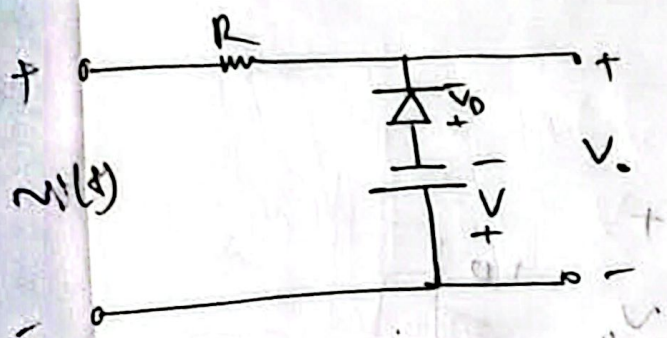
KVL in loop x,

$V_i - V_D - V = 0$

$\Rightarrow V_i = V - V_D$ = Transition level







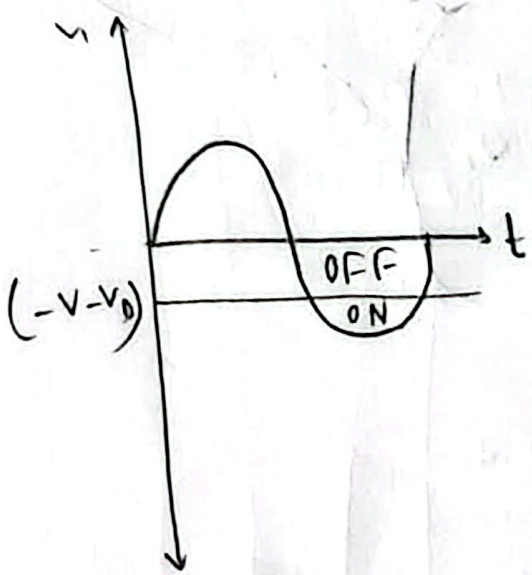
When diode is ON, $V_0 = -V - V_D$

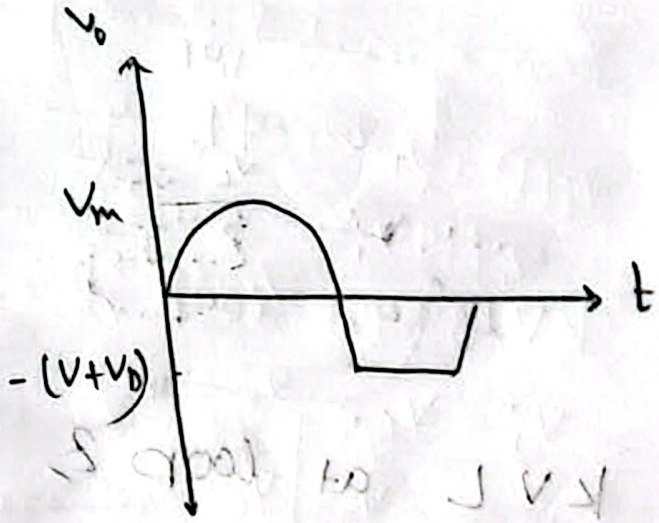
OFF, $V_0 = V_i$

KVL in loop x,

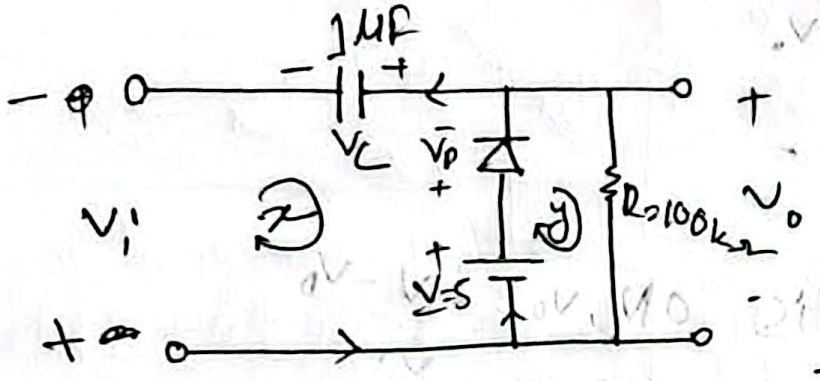
$V_i + V_D + V = 0$

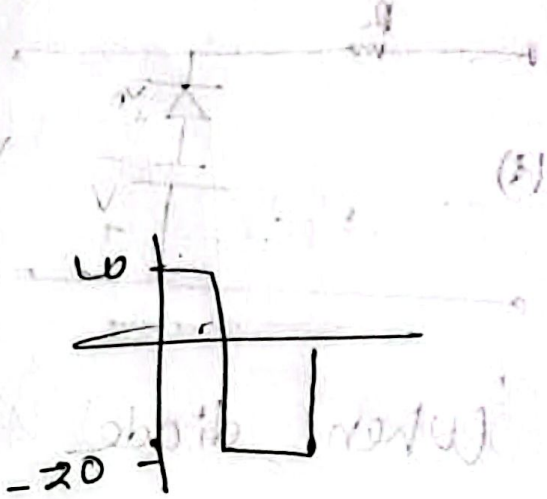
$\Rightarrow V_i = -V - V_D$ = Transition level.





Clamper:





KVL at loop 2,

$-V_i + V_C + V_D - V = 0$

$\Rightarrow -20 + V_C = V_i - V_D + V$

$\Rightarrow V_C = -20 - 0.7 + 5$

$= 24.3$

KVL at loop 3

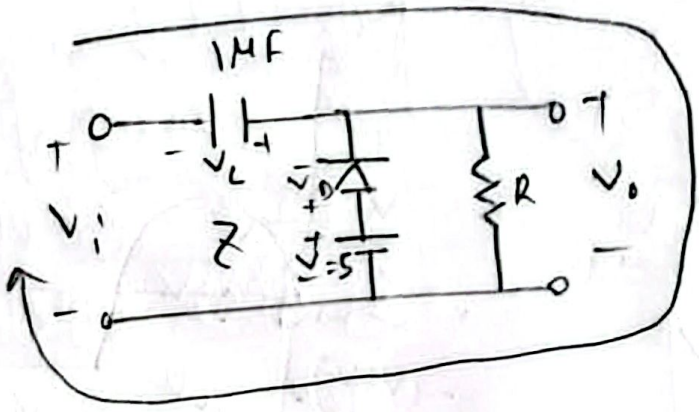
$-V_o + V - V_D = 0$

$\Rightarrow V_o = V - V_D$

$= 5 - 0.7$

$= 4.3V$

(For Negative half)



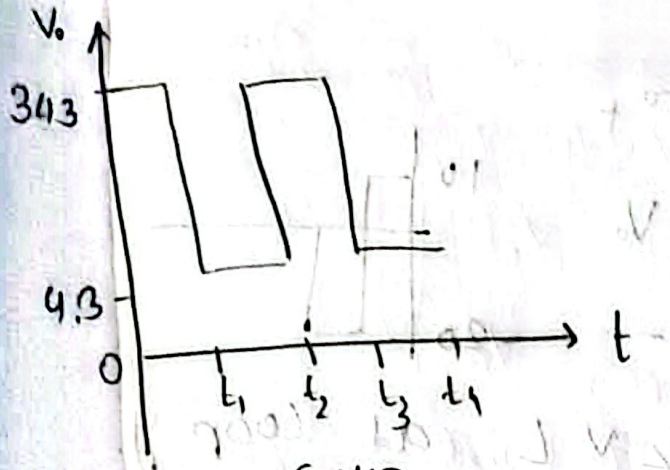
KVL at loop 2

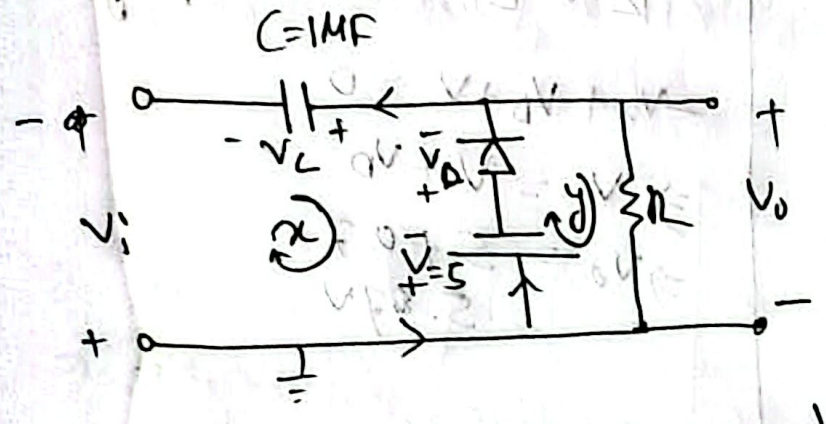
$V_i + V_C - V_o = 0$

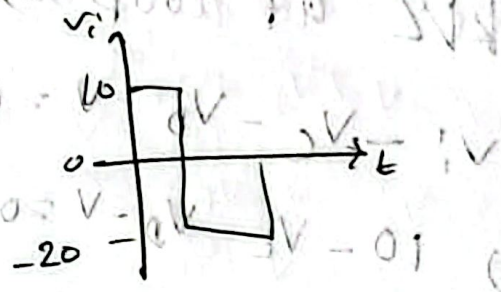
$\Rightarrow V_o = V_C + V_i$ (+ve half)

$= 24.3 + 10$

$= 34.3\ V$







KVL in loop x,

$-V_i + V_C + V_D + V = 0$

$\Rightarrow -20 + V_C + V_D + V = 0$

$\Rightarrow V_C = 20 - 0.7 - 5$

$= 14.3V$

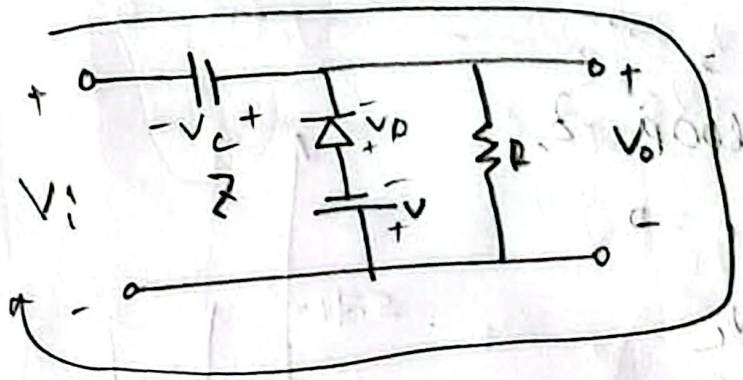
KVL in loop y,

$-V_o - V - V_D = 0$

$\Rightarrow V_o = -V - V_D$

$= -5 - 0.7$

$= -5.7V$



KVL at loop z,

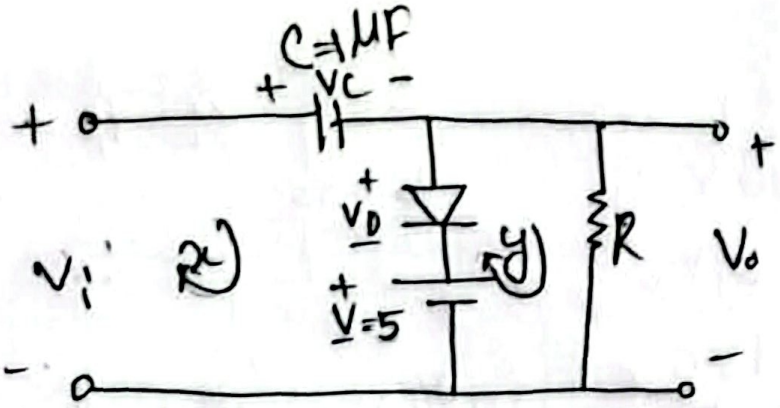
$V_i + V_C - V_o = 0$

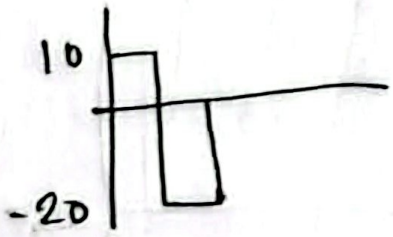
$\Rightarrow V_o = V_i + V_C$

$= 10 + 14.3V$

$= 24.3V$







KVL at loop x,

$V_i - V_C - V_D - V = 0$

$\Rightarrow 10 - V_C - V_D - V = 0$

$\Rightarrow V_C = 10 - 0.7 - 5 = 0$

$\Rightarrow V_C = 4.3V$

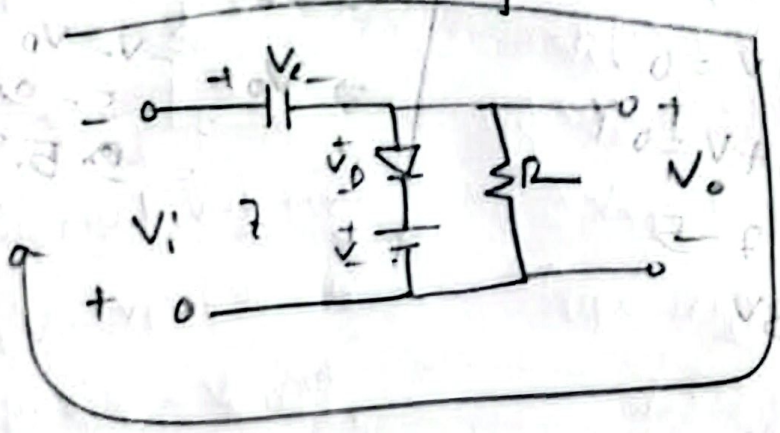
KVL at loop y.

$-V_o + V_D + V = 0$

$\Rightarrow V_o = V + V_D$

$\Rightarrow V_o = 5 + 0.7$

$= 5.07V$



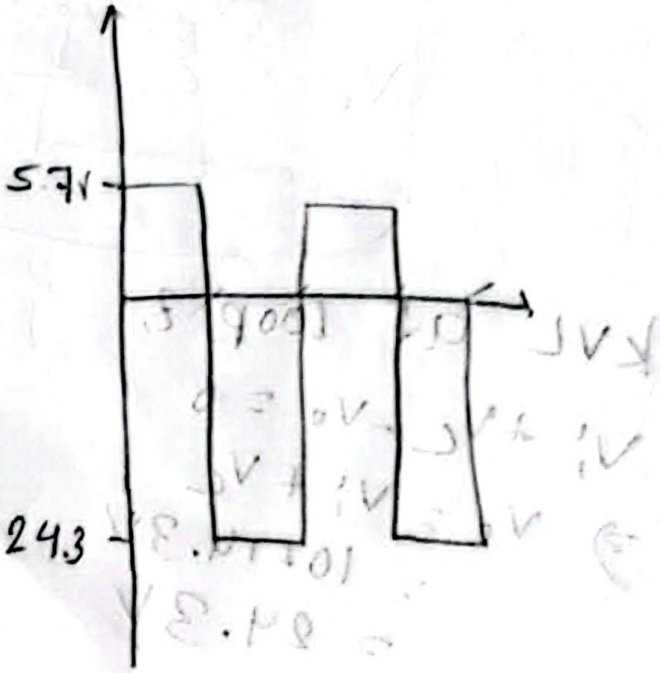
KVL at loop z,

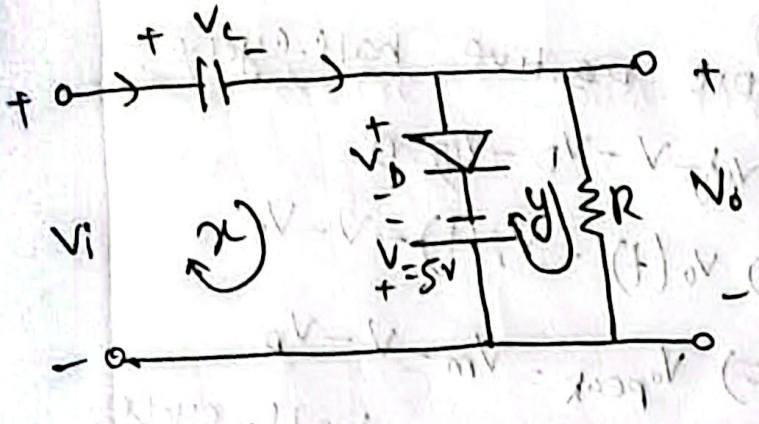
$-V_i - V_C - V_o = 0$

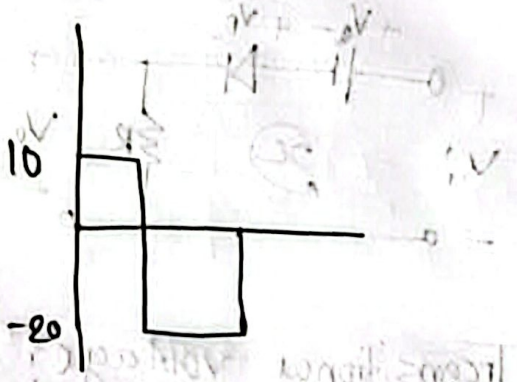
$\Rightarrow V_o = -V_i - V_C$

$= -20 - 4.3$

$= -24.3V$







KVL in loop x,

$V_i - V_C - V_D + V = 0$

$\Rightarrow 10 - V_C - 0.7 + 5 = 0$

$\Rightarrow V_C = 14.3\,V$

KVL in loop y,

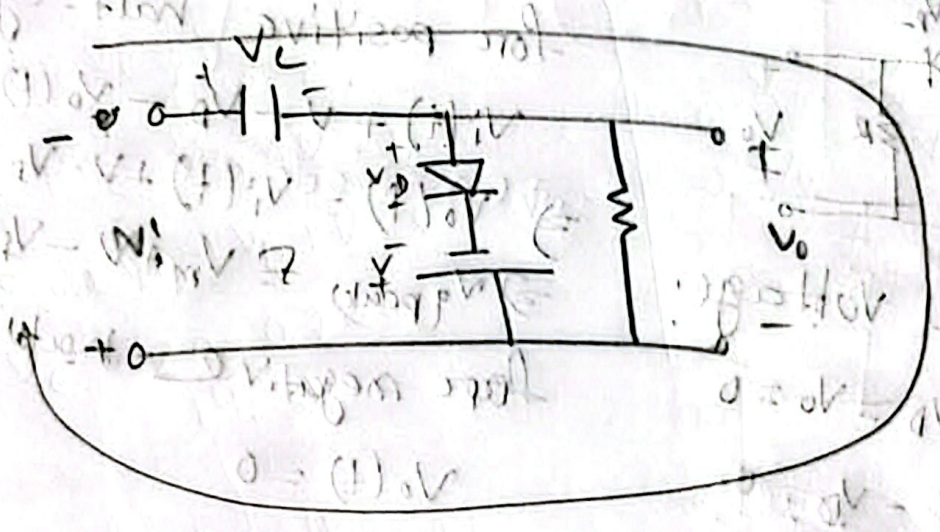
$-V_o - V + V_D = 0$

$\Rightarrow V_o = -V + V_D$

$= -5 + 0.7$

~~$= 4.3\,V$~~

$= -4.3\,V$



KVL in loop 2,

$-V_i - V_C - V_o = 0$

$\Rightarrow V_o = -V_i - V_C$

$= -20 - 14.3$

~~$= 24.3\,V$~~

$= -34.3\,V$

